

Master Benchmark and Empirical Accent Results Ledger

Authoritative Cross-Model Performance, Literature Reconciliation, and Demographic Subgroup Analysis

- **Project Title:** Disparity-Aware Continual Test-Time Adaptation for Accent-Robust Automatic Speech Recognition (DSG-CTTA)
- **Protocol Version:** `v1.0-cv27-amended` (Stage 0 Protocol Freeze & ADR-005)
- **Holdout Evaluation Stream:** Mozilla Common Voice 27.0 ($N = 900$ clips, 60 speakers, 6 strata, exactly 8,667 reference words)
- **Sentinel Panel:** Frozen External Calibration Panel ($M = 300$ clips, 30 disjoint speakers, 2,925 words)
- **Document Purpose:** Consolidated single-file empirical ledger for academic supervision, viva defense, and peer review.

1. Grounding & Denominator Invariants (Proof of Authenticity)

To verify that all reported error rates are mathematically authentic and reproducible, the entire evaluation is grounded on a pre-registered, frozen evaluation split (`datasets/splits/stage5_external_eval.csv`).

Accent Stratum	Total Clips	Distinct Speakers	Clips / Speaker	Reference Word Count (N_{ref})	Proportion of Corpus
US English	150	10	15	1,468 words	16.94%
England English	150	10	15	1,393 words	16.07%
South Asian English	150	10	15	1,430 words	16.50%
Australian English	150	10	15	1,493 words	17.23%
Canadian English	150	10	15	1,500 words	17.31%
Irish English	150	10	15	1,383 words	15.96%
TOTAL / SUM INVARIANT	900	60	15	8,667 words	100.00%

Sum Invariant: $1468 + 1393 + 1430 + 1493 + 1500 + 1383 = 8,667$ reference words

2. Table 1: Master Cross-Backbone Benchmark & Official Literature Reconciliation

This table reconciles published official benchmarks from original literature citations with our measured results across all 8 models.

Model Key	Architecture & Pretraining	Parameters	Published Benchmark Dataset	Published Benchmark WER	Official Citation	Measured Baseline No-Adapt	Measured SUTA	Measured DSUTA	Measured DMSUTA	Measured DSG-CTTA (Ours)	Net DSG vs Base
wav2vec2_base	CTC	94.4M	LibriSpeech test-clean / test-other (no LM)	3.4% / 8.6%	Baevski et al. (NeurIPS 2020), Table 1	22.45%	23.55%	22.55%	22.56%	22.52%	+0.07 pp
whisper_base	EncoderDecoder	72.6M	LibriSpeech test-other / Common Voice en zero-shot	10.5% / 13.1%	Radford et al. (ICML 2023), Table 8 & 9	14.32%	INCOMPATIBLE	INCOMPATIBLE	INCOMPATIBLE	INCOMPATIBLE	—
hubert_large	CTC	316.8M	LibriSpeech test-clean / test-other (no LM)	1.9% / 3.5%	Hsu et al. (IEEE/ACM TASLP 2021), Table 2	12.37%	12.77%	12.39%	12.26%	12.38%	+0.01 pp
data2vec_base	CTC	94.4M	LibriSpeech test-clean / test-other (no LM)	3.4% / 8.0%	Baevski et al. (ICML 2022), Table 1	20.43%	20.65%	19.79%	20.26%	20.32%	-0.12 pp
distil_whisper_small	EncoderDecoder	166.1M	Out-of-Distribution Average WER across 4 diverse datasets	10.3%	Gandhi et al. (EMNLP 2023), Table 1 & Model Card	9.18%	INCOMPATIBLE	INCOMPATIBLE	INCOMPATIBLE	INCOMPATIBLE	—
xlsr_english	CTC	315.5M	Common Voice 6.1 English Test Split	13.52%	Model Card: jonatasgrosman/wav2vec2-large-xlsr-53-english	11.81%	14.54%	11.88%	11.81%	11.81%	+0.00 pp
wav2vec2_large_lv60	CTC	315.5M	LibriSpeech test-clean / test-other (no LM)	1.9% / 4.1%	Baevski et al. (NeurIPS 2020), Table 1	12.93%	13.97%	13.00%	13.06%	12.91%	-0.02 pp
wav2vec2_large_robust	CTC	315.5M	LibriSpeech test-clean / test-other (no LM)	1.8% / 3.7%	Hsu et al. (Interspeech 2021) Robust Wav2Vec 2.0, Table 1	12.91%	13.28%	12.86%	12.92%	12.89%	-0.02 pp

3. Table 2: Complete Accent-by-Accent Performance Master Matrix

This table details the exact Word Error Rate (WER) achieved across each individual accent stratum for every model and adaptation method.

Model Key	Adaptation Method	US English (1,468 w)	England English (1,393 w)	South Asian English (1,430 w)	Australian English (1,493 w)	Canadian English (1,500 w)	Irish English (1,383 w)	Aggregate Stream WER (8,667 w)	Disparity Range (D)	Total Errors (E)
wav2vec2_base	No-Adapt Baseline	21.46%	20.32%	42.45%	22.91%	12.20%	15.62%	22.45%	30.25 pp	1,946 words
wav2vec2_base	SUTA (Unconstrained)	21.32%	21.82%	42.66%	24.05%	13.73%	18.00%	23.55%	28.92 pp	2,041 words
wav2vec2_base	DSUTA (Entropy Resets)	21.59%	20.17%	41.96%	22.97%	13.20%	15.55%	22.55%	28.76 pp	1,954 words
wav2vec2_base	DMSUTA (Memory Banks)	21.66%	20.60%	42.31%	22.44%	12.87%	15.69%	22.56%	29.44 pp	1,955 words
wav2vec2_base	DSG-CTTA (Ours)	21.53%	20.46%	42.17%	23.17%	12.67%	15.33%	22.52%	29.50 pp	1,952 words
hubert_large	No-Adapt Baseline	13.15%	11.27%	22.31%	11.05%	9.07%	7.38%	12.37%	14.93 pp	1,072 words
hubert_large	SUTA (Unconstrained)	13.28%	11.41%	23.08%	11.25%	9.47%	8.17%	12.77%	14.91 pp	1,107 words
hubert_large	DSUTA (Entropy Resets)	13.08%	11.49%	22.66%	10.72%	9.20%	7.23%	12.39%	15.43 pp	1,074 words
hubert_large	DMSUTA (Memory Banks)	13.08%	11.27%	22.38%	10.65%	9.13%	7.09%	12.26%	15.29 pp	1,063 words
hubert_large	DSG-CTTA (Ours)	12.94%	11.63%	22.66%	10.78%	9.00%	7.30%	12.38%	15.35 pp	1,073 words
data2vec_base	No-Adapt Baseline	20.37%	20.24%	38.46%	20.29%	11.33%	12.08%	20.43%	27.13 pp	1,771 words
data2vec_base	SUTA (Unconstrained)	20.30%	19.89%	36.85%	20.90%	12.40%	13.74%	20.65%	24.45 pp	1,790 words
data2vec_base	DSUTA (Entropy Resets)	19.82%	19.96%	36.50%	19.42%	11.13%	12.08%	19.79%	25.37 pp	1,715 words
data2vec_base	DMSUTA (Memory Banks)	19.62%	20.24%	38.53%	20.03%	11.13%	12.22%	20.26%	27.40 pp	1,756 words
data2vec_base	DSG-CTTA (Ours)	20.37%	20.17%	37.90%	20.09%	11.40%	12.15%	20.32%	26.50 pp	1,761 words
xlsr_english	No-Adapt Baseline	13.76%	12.06%	16.99%	11.32%	8.07%	8.75%	11.81%	8.93 pp	1,024 words

Model Key	Adaptation Method	US English (1,468 w)	England English (1,393 w)	South Asian English (1,430 w)	Australian English (1,493 w)	Canadian English (1,500 w)	Irish English (1,383 w)	Aggregate Stream WER (8,667 w)	Disparity Range (D)	Total Errors (E)
xlsr_english	SUTA (Unconstrained)	13.42%	12.99%	18.39%	15.14%	12.53%	14.82%	14.54%	5.86 pp	1,260 words
xlsr_english	DSUTA (Entropy Resets)	13.49%	12.20%	16.92%	11.39%	8.27%	9.11%	11.88%	8.66 pp	1,030 words
xlsr_english	DMSUTA (Memory Banks)	13.76%	12.20%	16.78%	11.39%	8.07%	8.75%	11.81%	8.72 pp	1,024 words
xlsr_english	DSG-CTTA (Ours)	13.76%	12.06%	16.99%	11.32%	8.07%	8.75%	11.81%	8.93 pp	1,024 words
wav2vec2_large_lv60	No-Adapt Baseline	14.85%	12.35%	20.14%	12.46%	9.27%	8.53%	12.93%	11.61 pp	1,121 words
wav2vec2_large_lv60	SUTA (Unconstrained)	15.05%	12.28%	21.47%	13.26%	10.33%	11.50%	13.97%	11.14 pp	1,211 words
wav2vec2_large_lv60	DSUTA (Entropy Resets)	14.92%	12.35%	19.93%	12.53%	9.73%	8.53%	13.00%	11.40 pp	1,127 words
wav2vec2_large_lv60	DMSUTA (Memory Banks)	14.78%	12.49%	20.35%	12.53%	9.33%	8.89%	13.06%	11.46 pp	1,132 words
wav2vec2_large_lv60	DSG-CTTA (Ours)	14.85%	12.35%	20.14%	12.46%	9.13%	8.53%	12.91%	11.61 pp	1,119 words
wav2vec2_large_robust	No-Adapt Baseline	15.94%	12.85%	18.81%	11.65%	9.20%	9.04%	12.91%	9.77 pp	1,119 words
wav2vec2_large_robust	SUTA (Unconstrained)	15.46%	12.42%	18.60%	11.86%	9.93%	11.50%	13.28%	8.67 pp	1,151 words
wav2vec2_large_robust	DSUTA (Entropy Resets)	15.67%	12.56%	18.46%	12.06%	9.40%	9.04%	12.86%	9.42 pp	1,115 words
wav2vec2_large_robust	DMSUTA (Memory Banks)	15.46%	12.99%	18.39%	12.19%	9.47%	9.04%	12.92%	9.35 pp	1,120 words
wav2vec2_large_robust	DSG-CTTA (Ours)	15.87%	12.99%	18.60%	11.79%	9.13%	8.97%	12.89%	9.64 pp	1,117 words
whisper_base	No-Adapt Baseline	16.55%	14.43%	21.40%	13.53%	7.87%	12.36%	14.32%	13.53 pp	1,241 words
distil_whisper_small	No-Adapt Baseline	10.90%	9.62%	13.50%	8.24%	6.40%	6.51%	9.18%	7.10 pp	796 words

4. Table 3: Exact Edit Operation Breakdown (Substitutions, Deletions, Insertions)

Below are the exact raw edit distances computed via dynamic string alignment across all 6 accent strata for each model under baseline (No-Adapt):

Model Key	Accent Stratum	Reference Words	Substitutions (<i>S</i>)	Deletions (<i>D</i>)	Insertions (<i>I</i>)	Total Errors (<i>S</i> + <i>D</i> + <i>I</i>)	Measured WER
data2vec_base	US English	1468	231	34	34	299	20.37%
data2vec_base	England English	1393	235	21	26	282	20.24%
data2vec_base	South Asian English	1430	439	65	46	550	38.46%
data2vec_base	Australian English	1493	245	34	24	303	20.29%
data2vec_base	Canadian English	1500	141	12	17	170	11.33%
data2vec_base	Irish English	1383	142	12	13	167	12.08%
distil_whisper_small	US English	1468	124	16	20	160	10.90%
distil_whisper_small	England English	1393	100	11	23	134	9.62%
distil_whisper_small	South Asian English	1430	149	14	30	193	13.50%
distil_whisper_small	Australian English	1493	88	20	15	123	8.24%
distil_whisper_small	Canadian English	1500	79	10	7	96	6.40%
distil_whisper_small	Irish English	1383	72	12	6	90	6.51%
hubert_large	US English	1468	163	8	22	193	13.15%
hubert_large	England English	1393	127	11	19	157	11.27%
hubert_large	South Asian English	1430	269	25	25	319	22.31%
hubert_large	Australian English	1493	135	10	20	165	11.05%
hubert_large	Canadian English	1500	109	12	15	136	9.07%
hubert_large	Irish English	1383	86	8	8	102	7.38%
wav2vec2_base	Australian English	1493	271	32	39	342	22.91%
wav2vec2_base	Canadian English	1500	153	13	17	183	12.20%
wav2vec2_base	England English	1393	227	27	29	283	20.32%
wav2vec2_base	Irish English	1383	179	23	14	216	15.62%
wav2vec2_base	South Asian English	1430	500	53	54	607	42.45%
wav2vec2_base	US English	1468	253	30	32	315	21.46%
whisper_base	US English	1468	176	16	51	243	16.55%
whisper_base	England English	1393	151	12	38	201	14.43%

Model Key	Accent Stratum	Reference Words	Substitutions (<i>S</i>)	Deletions (<i>D</i>)	Insertions (<i>I</i>)	Total Errors (<i>S</i> + <i>D</i> + <i>I</i>)	Measured WER
whisper_base	South Asian English	1430	229	19	58	306	21.40%
whisper_base	Australian English	1493	153	21	28	202	13.53%
whisper_base	Canadian English	1500	98	9	11	118	7.87%
whisper_base	Irish English	1383	134	10	27	171	12.36%
xlsr_english	US English	1468	161	26	15	202	13.76%
xlsr_english	England English	1393	136	18	14	168	12.06%
xlsr_english	South Asian English	1430	203	25	15	243	16.99%
xlsr_english	Australian English	1493	136	21	12	169	11.32%
xlsr_english	Canadian English	1500	104	12	5	121	8.07%
xlsr_english	Irish English	1383	99	15	7	121	8.75%
wav2vec2_large_lv60	US English	1468	178	15	25	218	14.85%
wav2vec2_large_lv60	England English	1393	142	7	23	172	12.35%
wav2vec2_large_lv60	South Asian English	1430	241	26	21	288	20.14%
wav2vec2_large_lv60	Australian English	1493	152	15	19	186	12.46%
wav2vec2_large_lv60	Canadian English	1500	110	12	17	139	9.27%
wav2vec2_large_lv60	Irish English	1383	102	8	8	118	8.53%
wav2vec2_large_robust	US English	1468	197	17	20	234	15.94%
wav2vec2_large_robust	England English	1393	148	8	23	179	12.85%
wav2vec2_large_robust	South Asian English	1430	229	21	19	269	18.81%
wav2vec2_large_robust	Australian English	1493	144	13	17	174	11.65%
wav2vec2_large_robust	Canadian English	1500	117	9	12	138	9.20%
wav2vec2_large_robust	Irish English	1383	105	10	10	125	9.04%

5. Table 4: DSG Online Controller Decision & Safety Statistics

Across all 6 adaptable CTC architectures, the Disparity Safety Gate evaluated 1,350 candidate updates across 225 sequential stream windows:

Model Key	Model ID	Architecture	Candidate Updates	Accepted Updates	Rejected Updates	Acceptance Rate	Statistical Violations	Fail-Closed Errors	Final Net Δ_R vs Base	Worst Subgroup Harm ($\max_g \Delta_g$)
wav2vec2_base	facebook/wav2vec2-base-960h	CTC	225	9.0	216.0	4.0%	216.0	0.0	+0.07 pp	+0.47 pp
whisper_base	openai/whisper-base	EncoderDecoder	INCOMPATIBLE	nan	nan	nan	nan	nan	+nan pp	+nan pp
hubert_large	facebook/hubert-large-ls960-ft	CTC	225	6.0	219.0	2.7%	219.0	0.0	+0.01 pp	+0.36 pp
data2vec_base	facebook/data2vec-audio-base-960h	CTC	225	1.0	224.0	0.4%	224.0	0.0	-0.12 pp	+0.07 pp
distil_whisper_small	distil-whisper/distil-small.en	EncoderDecoder	INCOMPATIBLE	nan	nan	nan	nan	nan	+nan pp	+nan pp
xlsr_english	jonatasmrosman/wav2vec2-large-xlsr-53-english	CTC	225	0.0	225.0	0.0%	225.0	0.0	+0.00 pp	+0.00 pp
wav2vec2_large_lv60	facebook/wav2vec2-large-960h-lv60	CTC	225	2.0	223.0	0.9%	223.0	0.0	-0.02 pp	+0.00 pp
wav2vec2_large_robust	facebook/wav2vec2-large-robust-ft-libri-960h	CTC	225	5.0	220.0	2.2%	220.0	0.0	-0.02 pp	+0.14 pp

6. Scientific Defense & Authenticity Reconciliation for Professors and Reviewers

When presenting these empirical numbers, reviewers often ask two key questions:

Q1: Why are the Common Voice error rates (11% - 42%) higher than published LibriSpeech benchmarks (2% - 8%)?

- Acoustic Domain Divergence:** LibriSpeech consists of clean, professional audiobooks read by native North American speakers in acoustically controlled environments using studio-grade condenser microphones.
- Demographic and Accent Shift:** Mozilla Common Voice 27.0 captures crowdsourced recordings from real-world laptop and smartphone microphones across global English speakers. For example, South Asian English exhibits phonemic retroflexion and syllable timing that diverges sharply from North American training priors (resulting in 42.45% baseline WER on wav2vec2_base).
- Consistency with Literature:** This divergence is fully corroborated by published literature: Koenecke et al. (PNAS 2020) demonstrated a 2x error rate elevation across diverse speakers; Radford et al. (ICML 2023) showed that Whisper error rates jump from 2.7% on LibriSpeech clean to 13.1% on global Common Voice English.

Q2: How can the results be verified independently?

Every single number in this report is backed by raw row-level prediction CSVs stored in `reports/stage6/checkpoints/`: - Every transcribed audio clip has its exact audio ID, speaker ID, accent code, normalized reference text, model hypothesis string, and exact execution timestamp. - Any independent auditor can open any CSV in Excel or Python and run `(sub + del + ins) / ref_words` to verify bit-for-bit numerical reproducibility.

7. Artifact File Directory Reference

- **Master CSV Matrix:** [reports/stage6/master_model_accent_benchmark_matrix.csv](#)
- **8-Model Overall CSV:** [reports/stage6/eight_model_benchmark.csv](#)
- **Complete Accent CSV:** [reports/stage6/eight_model_group_metrics.csv](#)
- **Dissertation Dossier PDF:** [reports/project_report/master_research_report.pdf](#)
- **Formula Reference PDF:** [reports/project_report/formula_reference.pdf](#)